



Proline: The Natural Armor of Fruit Crops

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Abstract

Abiotic stresses such as extreme temperatures, salinity, heavy metals, nutrient deficiencies, water logging and chilling affect fruit crops throughout their life cycle, causing cellular changes like protein denaturation, membrane injury, reactive species accumulation and osmotic stress. This ultimately reduces fruit yield and quality. Under such conditions, plants accumulate osmolytes to mitigate stress, among which proline is one of the most effective. This proteinogenic amino acid plays key roles in plant growth and stress tolerance. However, its synthesis may decline under severe stress, making exogenous application a suitable strategy to enhance proline levels and improve stress resilience.

Introduction

Fruit growing is one of the most important and age-old tradition practised in India, contributing significantly to human welfare and national economy. With diverse climates, India ranks second in global fruit production at 112.73 million metric tonnes. However, climate change, natural disasters, urbanization and industrialization expose fruit crops to abiotic stresses which limit growth, reduce productivity, quality of fruit crops. Several different attempts are being made to minimize the negative effects of these stresses like chemical fertilization, genetic engineering, breeding programmes and plant stimulation by exogenous application of molecules to plants. Chemical fertilization is becoming a major threat because of its negative impact on human health, hence in the recent year application of Phyto protectants found effective in mitigating the damages caused by stress in plants.

Types of Phyto protectants

- Microbial origin Phyto protectants - Arbuscular mycorrhiza and Rhizobacteria
- N- containing Phyto protectants - Osmoprotectants (amino acids, betains, polyamines)
- Phytohormones - Absciscic acid, Jasmonic acid, Salicylic acid and Brassinosteroides

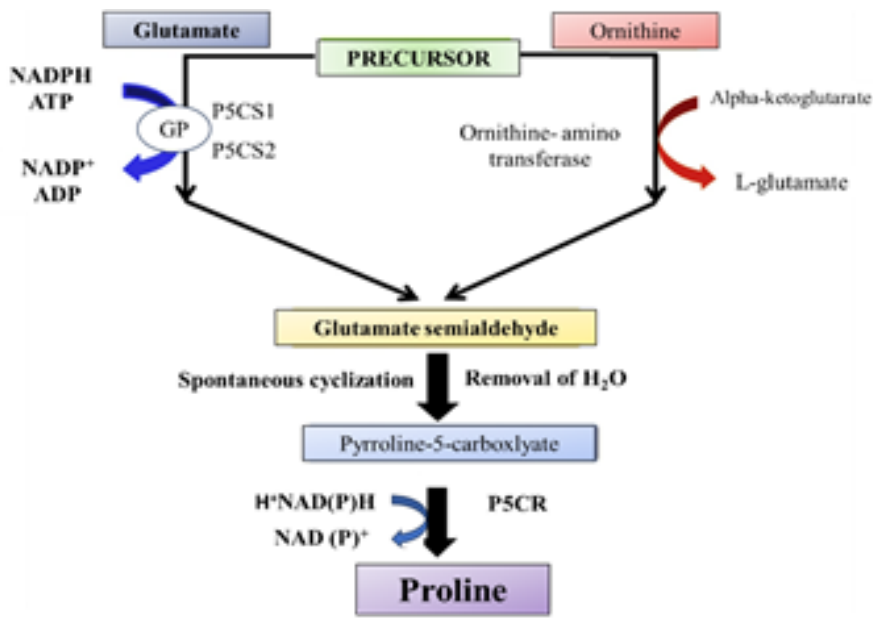
Among this application of Osmoprotectants has attracted a lot due to its high efficiency, ease of use, low cost, no need of advanced equipment, environmentally safer and immediate action. One such most effective Osmoprotectant is proline which we do not hear in our day-to-day life but quietly works behind the scene in order to overcome stress in plants.

Proline

It is an aliphatic, non-polar amino acid that serves as a vital osmoregulator in plants that are subjected to abiotic stresses and is endogenously produced in plant to enhance its ability to withstand adverse environmental conditions. It stabilizes the cell wall structure and maintains the cellular turgor which is crucial for maintaining fruit quality under tough conditions.

Proline’s protective role under abiotic stress

- Osmoprotection
- Stabilizes protein structure
- Act as metal chelator
- Scavenges reactive oxygen species
- Act as signaling molecule-
- Redox buffer reductant



Biosynthesis of proline

In order to overcome these kinds of stresses, proline is synthesized in plants by 2 pathways.

- Glutamate pathway
- Ornithine pathway

Glutamate pathway is catalysed by two enzymes P5CS and P5CR. Glutamate is first phosphorylated by glutamyl phosphate (GP), then the enzyme Δ^1 -pyrroline-5-carboxylate synthetase (P5CS) converts it into an intermediate compound glutamic semialdehyde (GSA). This GSA through continuous cyclization gets converted into Δ^1 -pyrroline-5-carboxylate (P5C) which is then with the help of enzyme Δ^1 -pyrroline-5-carboxylate reductase (P5CR) reduces into proline.

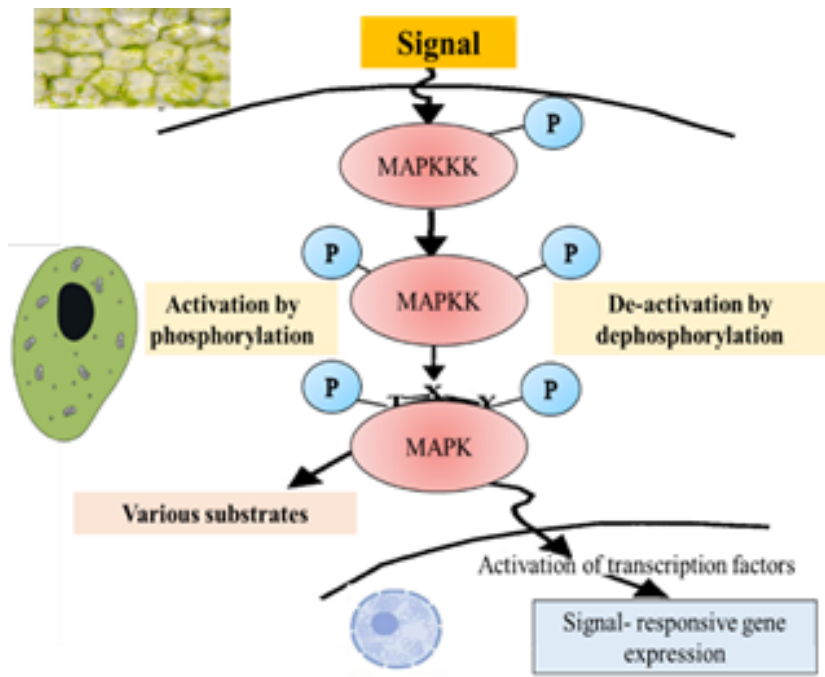
Proline can be synthesised by an alternative pathway through ornithine. Ornithine is transaminated into GSA which is catalysed by the enzyme ornithine amino transferase (OAT) and converted to proline through P5CS enzyme.

Metabolism of proline

Elijah Adams and Harold Strecker first outlined the proline metabolic pathway in the 1950s. Proline is synthesized in the cytosol and chloroplast via the glutamate and ornithine pathways, while its catabolism occurs in mitochondria through the proline cycle. The ERD5 gene encodes proline dehydrogenase (PDH), which converts proline to P5C, later transformed into glutamate by ProDH in the inner mitochondrial membrane. Proline degradation increases when its levels are excessive or no longer needed. Under stress, plants boost proline accumulation by upregulating P5CS and downregulating PDH.

Signal transduction

When cells perceive stress, signals reach the cytoplasm and activate the MAPK cascade, which regulates antioxidant pathways. This cascade includes MAPKKKs, MAPKKs and MAPKs. MAPKs contain a conserved T-X-Y motif in their activation loop, where dual phosphorylation of threonine and tyrosine by MAPKKs activates them. The activated MAPKs then phosphorylate downstream substrates and transcription factors in the nucleus, promoting the expression of genes involved in proline biosynthesis.



TRANSPORTATION

Proline transport is essential for helping plants cope with stress. It moves across the plasma membrane through AAP and APC transporter families, with ProT transporters enabling long-distance movement. Glutamine synthetase also regulates proline distribution in tissues such as xylem and phloem. Synthesized proline is loaded into phloem via ProT transporters and delivered to growing organs, where it functions as an osmoprotectant, nitrogen reserve and energy precursor.

Regulatory factors and their gene expression in regarding proline accumulation, biosynthesis under different environmental stress conditions.

- ALA- Up- regulates the P5CS and downregulates the proDH expression
- Gene OsNAC5- Increases the proline accumulation
- SNP- Enhances the P5CS activity and reduces the PDH activity
- 24-Epibrassinolide- Increases the proline accumulation
- High expression of KvP5CS1 and KvOAT, low expression of KvPDH- Increases the proline accumulation under salt stress
- Over expression of bHLH protein- Enhances the proline accumulation under osmotic condition
- Over expression P5CS1 and P5CR- Increases the biosynthesis of proline under drought stress

Proline mode of action in enhancing fruit quality

1.Through osmoregulation

Under heat, salinity and water stress, osmoregulation is essential for maintaining cellular integrity. Increased proline lowers osmotic potential, preserves turgor and supports water uptake, enabling normal

cellular function and promoting fruit growth even under limited moisture. Proline also stabilizes proteins and cellular structures, protecting them from dehydration and salt-induced damage, which ultimately enhances fruit quality and firmness.

2. Through antioxidant capacity

Proline plays a key role in protecting plants from oxidative stress by directly scavenging reactive oxygen species (ROS) such as superoxide radicals and hydrogen peroxide, thereby preventing damage to proteins, lipids and nucleic acids. It also enhances the activity of major antioxidant enzymes like catalase, peroxidases and superoxide dismutase, strengthening the plant’s overall defence system. This improved antioxidant protection helps maintain cellular integrity and supports better fruit firmness, colour and quality. Additionally, proline contributes to cellular redox balance, optimizing metabolic processes that promote fruit development, extended shelf life and improved nutritional value.

3. Through photosynthetic activity

Proline protect and stabilize the photosynthetic apparatus by preserving chloroplast structure and preventing degradation of photosystem complexes, ensuring steady photosynthetic activity. It also influences stomatal conductance, reducing water loss while improving CO₂ uptake, allowing photosynthesis to continue under adverse conditions. Proline application has been linked to increased chlorophyll content, enhancing light absorption and overall photosynthetic efficiency. Together, these effects boost fruit productivity, leading to larger fruits, higher yields and improved quality traits such as taste, colour and nutritional value.

Exogenous application of proline

When proline is applied externally to stressed plants, receptor-mediated signal transduction activates the plant’s defense system, enhancing the synthesis of metabolites, osmoprotectants and antioxidant enzymes. This upregulates endogenous proline and antioxidant-related genes, increasing the activity of superoxide dismutase, ascorbate peroxidase, catalase and glutamate reductase. These enzymes scavenge free radicals, stabilize membranes, maintain redox balance and improve photosynthesis, ultimately enhancing plant stress tolerance.

Products of proline



Conclusion

Proline a naturally occurring amino acid can mitigate the damages caused by environmental stresses. It affects plant growth and production through metabolic signal mediating function and regulate expression of enzymes involved in proline synthesis which will be useful for improving the stress tolerance in plants. Exogenous application of proline as a stress mitigator has been revealed to be a most promising strategy for enhancing the resilience of fruit crops to stress and improve their productivity and quality without leaving any chemical residues.

References

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